CIS 371 Web Application Programming

Vuetify II

GRAND VALLEY STATE UNIVERSITY

Lecturer: Dr. Haoyu Li

Based on the original version by Professor Hans Dulimarta and Professor Yong Zhuang.

Exercise from the previous class

Practice: User Registration Form

Requirements

- Add a v-text-field for the user's name.
- Add a v-select for the user's preferred programming language, with options:
  - JavaScript
  - Python
  - Java
- Add a v-radio-group to let the user choose their experience level:
  - Beginner
  - Intermediate
  - Advanced

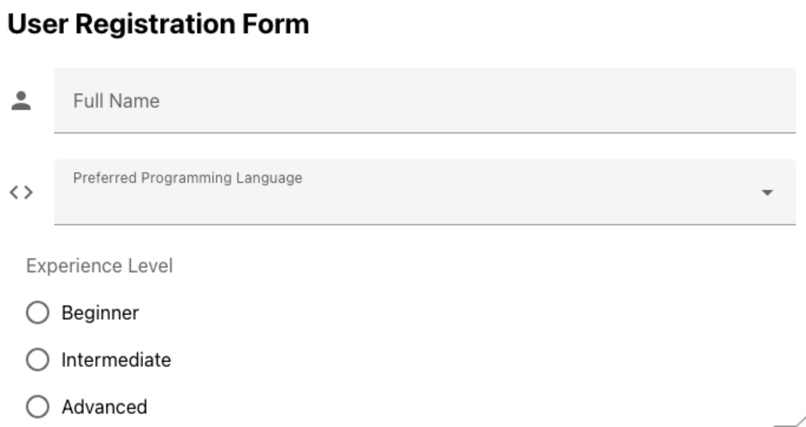
Answer

Practice

```
<!-- 1. Add a text field for the user's name -->
<!-- TODO -->
<v-text-field
v-model="name"
label="Full Name"
prepend-icon="mdi-account"
required
></v-text-field>
```

```
<!-- 2. Add a select field for preferred programming language -->
<!-- TODO -->
<v-select
v-model="language"
:items="languages"
label="Preferred Programming Language"
prepend-icon="mdi-code-tags"
required
></v-select>
```

```
<v-radio-group v-model="level" label="Experience Level">
<v-radio
v-for="(l,idx) in levels"
:label="l"
:value="l"
:key="idx"
></v-radio>
</v-radio-group>
{{level}}
```



Grid System

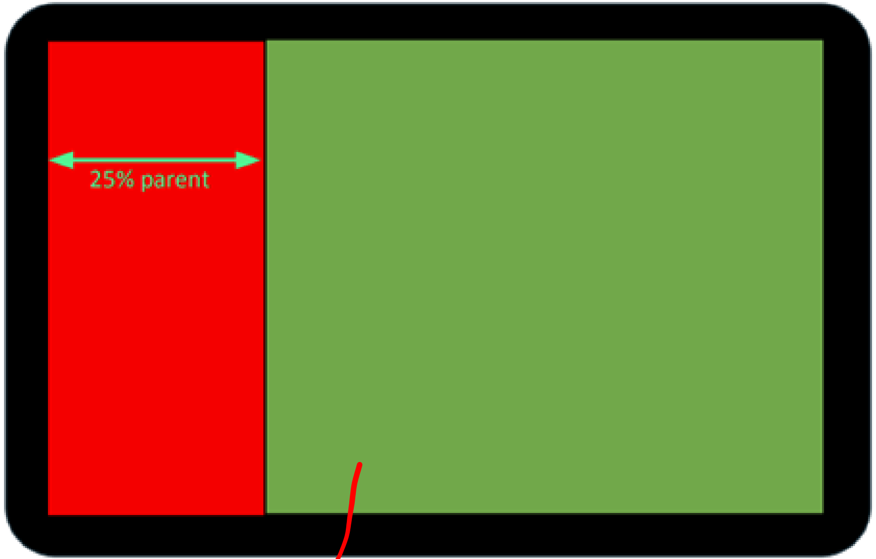
Vuetify Grid System

- Inspired by Bootstrap grid & CSS Flexbox
- Relevant elements: v-row, v-col,
  - Warning: row/col does not imply “horizontal”/”vertical” grouping of elements
- Vuetify 12-column layout?
  - Why 12? It is a relatively small number with many integer factors: 2, 3, 4, 6
  - It is easy to divide the screen into
    - Full width (100%)
    - 2 halves (each column is 50% of total screen width => 6/12 each)
    - 3 thirds (33% -> 4/12 each)
    - 4 quarters (25% => 3/12 each)

Demo

Vuetify Grid Layout (12-column system)

```
<v-container>
  <v-row>
    <v-col cols="3" class="bg-red"> </v-col>
    <v-col class="bg-green"> </v-col>
  </v-row>
</v-container>
```



Users may use your web app on a variety of screen sizes.

- Smartphones, tablet, desktop, HDTV, Ultra HDTV, …

Can't use one-layout-fits-all approach!

Responsive Layout

Traditional Solution: CSS @media query

- @media (max-width: 600px) { visual styles #1 }
- @media (max-width: 1024px) { visual styles #2 }
- @media (max-width: 1880px) { visual styles #3 }

Vuetify Solution

- Use **relative dimensions** based on 12-column grid
- Use **fluid** container (as opposed to fixed size)
- Size code & media **breakpoints**

Size Code and Media breakpoints

Symbolic names for sizes based on 12-column grid layout

Code	Description	Media	Size Range
xs	eXtra Small	Small to large phone	up to 600 px
sm	Small	Small to medium tablet	600 - 840 px
md	Medium	Large tablet to laptop	840 - 1145 px
lg	Large	Laptop to desktop	1145 - 1545 px
xl	Extra Large	1080p to 1440p desktop	1545 - 2138 px
xxl	Extra extra Large	4k and ultra-wide	> 2138 px

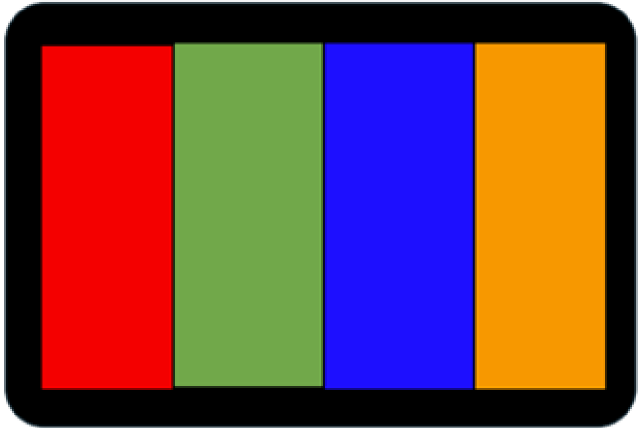
Examples: Cols

- xs=3 ⇒ **25% width by default**, from extra-small screens upward **until overridden**
- sm=6 ⇒ **50% width on small screens and above**, until overridden by md
- md=12 ⇒ **100% width on medium screens and above**, until overridden by lg
- lg=9 ⇒ **75% width on large screens and above**, until overridden by xl
- xl=2 ⇒ **16.67% width on extra-large screens and above**

Medium vs. Small

```
<v-container>
  <v-row>
    <v-col md="3" class="red" sm="6"></v-col>
    <v-col md="3" class="green" sm="6"></v-col>
    <v-col md="3" class="blue" sm="6"></v-col>
    <v-col md="3" class="orange" sm="6"></v-col>
  </v-row>
</v-container>
```

On medium screens



On small screens

Size Code and Media breakpoints

```
<v-container>
  <v-row>
    <v-col v-for="url in imageURLs" lg="2" md="3" sm="4" xs="6">
      <img :src="url" />
    </v-col>
  </v-row>
</v-container>
```

Screen Size	Effective Width	# of images per line?
Large	2	
Medium	3	
Small	4	
Extra Small	6	

Demo

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Size Code and Media breakpoints

```
<v-container>
  <v-row>
    <v-col v-for="url in imageURLs" lg="2" md="3" sm="4" xs="6">
      <img :src="url" />
    </v-col>
  </v-row>
</v-container>
```

Screen Size	Effective Width	# of images per line?
Large	2	6
Medium	3	4
Small	4	3
Extra Small	6	2

Demo

Styling via Class Names

Vuetify provides predefined class names for styling the following properties:

- Material Color classes (background and foreground)
- Media size breakpoints
- Space classes (margin & padding)

Styling Margin/Padding via Class Names

- Vuetify provides predefined classnames (zz-n) as shortcuts for defining margins and paddings
- Regex for these class names:

[mpg] [tblrseaxyc] - [0-16 | n1-16 | auto]

margin / padding / gap

top / bottom / left / right / start / end / all
x (left&right) / y (top&bottom) / column-gap

[Vuetify Docs - Spacing]()

Styling Margin/Padding via Class Names

[mpg] [tblrseaxyc] - [0-16 | n1-16 | auto]

Code	Pixels
0	0
1	4px
2	8px
3	12px
4	16px
n1	-4px
n3	-12px

negative number only work for margin.

Vuetify CSS Class	Equivalent CSS Declaration
ma-2	
ml-3	
pt-0	
pb-5	
px-1	
me-n2	

Spacing Class Names

```html
<!-- traditional HTML file --->
<span id="msg">Sample Text</span>

<style>
  #msg {
    margin-left: 24px;
  }
</style>
```

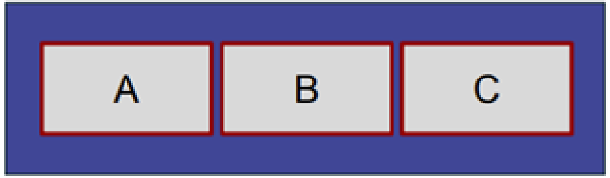
Traditional way

```html
<span class="ml-4">Sample Text with left margin</span>
<p class="py-2">text with top and bottom padding…</p>
```

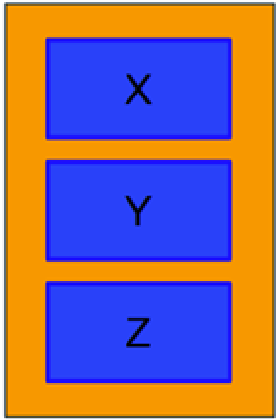
The vuetify way

Row- vs. Column-Oriented Box

```
<v-container>
  <v-row>
    <v-col>A</v-col>
    <v-col>B</v-col>
    <v-col>C</v-col>
  </v-row>
</v-container>
```




```
<v-container>
  <v-row class="flex-column">
    <v-col>X</v-col>
    <v-col>Y</v-col>
    <v-col>Z</v-col>
  </v-row>
</v-container>
```



v-row: justify-* and align-*

CSS3 FlexBox	CSS3 Flexbox Properties	Vuetify Attribute

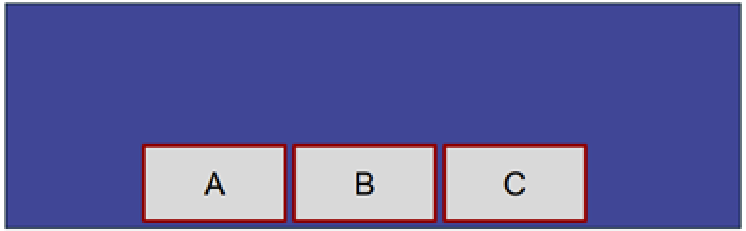
Main-axis ***justification***	`justify-content: flex-start;`
	`justify-content: flex-end;`
	`justify-content: flex-center;`
	`justify-content: flex-space-around;`
	`justify-content: flex-space-between;`
		`justify=”start”`
		`justify=”end”`
		`justify=”center”`
		same as above

Cross-axis ***alignment***	`align-items: flex-start;`
	`align-items: flex-end;`
	`align-items: flex-center;`
	`align-items: flex-baseline;`
		`align=”start”`
		`align=”end”`
		same as above

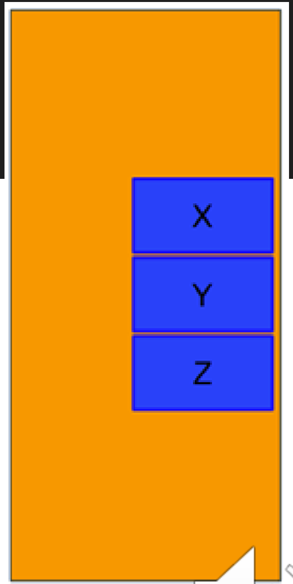
Vuetify accepts both align or align-content as valid attributes

Justification vs. Alignment

```
<v-container>
  <v-row justify="center" align="end">
    <v-col>A</v-col>
    <v-col>B</v-col>
    <v-col>C</v-col>
  </v-row>
</v-container>
```




```
<v-container>
  <v-row class="flex-column" justify="center" align="end">
    <v-col>X</v-col>
    <v-col>Y</v-col>
    <v-col>Z</v-col>
  </v-row>
</v-container>
```



Components

- Navigation: App Bars, Navigation Drawers, Footer Component, System Bar, Tabs Components, …,
- Containment: Bottom Sheet component, Card Components, Button Component, List Components, Chip Component, Expansion Panel Components, Dialog Component, Toolbar Components, …,
- Form Inputs and controls: Autocomplete Component, Combobox Components, Text Field, Checkbox Components, Radio Button, Number Input Component …
- Layouts: Grid System
- Selection: Carousel Component, Window Components, Stepper Components, Chip Group, Button Group.
- Data and Display: Confirm Edit Component, Data Iterator Component, Data Table Component, Sparkline Component, …,
- Feedback: Alert Component, Badge Component, Empty State Component, Skeleton loader component, Progress Component, Rating Component, …,
- Images, Icons, Pickers, …,
- …

Online Doc

Containment: Bottom sheet

```
<template>
  <div class="text-center">
    <v-btn size="x-large" text="Click Me" @click="sheet = !sheet"></v-btn>

    <v-bottom-sheet v-model="sheet" inset>
      <v-card class="text-center" height="200">
        <v-card-text>
          <v-btn variant="text" @click="sheet = !sheet"> close </v-btn>
          <div>This is a bottom sheet that is using the inset prop</div>
        </v-card-text>
      </v-card>
    </v-bottom-sheet>
  </div>
</template>

<script lang="ts" setup>
import { ref } from "vue";
const sheet = ref(false);
</script>
```

With the inset prop, reduce the maximum width of the content area on desktop to 70%.

```
<template>
  <v-app>
    <v-container>
      <v-bottom-sheet>
        <template v-slot:activator="{ props }">
          <v-btn v-bind="props" text="Click Me"></v-btn>
        </template>
        <v-card
          title="Bottom Sheet"
          text="Lorem ipsum dolor sit amet consectetur,..."
        ></v-card>
      </v-bottom-sheet>
    </v-container>
  </v-app>
</template>
```

[Demo]

[Vuetify Docs]

[Demo]

Containment: Buttons

Vuetify Docs

```
<v-btn icon="mdi-account" size="x-small"></v-btn>
```

The **size** property is used to control the size of the button and scales with density.

```
<v-btn density="default" icon="mdi-open-in-new"></v-btn>
```

The **density** prop is used to control the vertical space that the button takes up.

```
<v-btn append-icon="mdi-account" size="x-small">Tom</v-btn>
```

```
<v-btn prepend-icon="mdi-account" size="x-small">Tom</v-btn>
```

Props like prepend-icon and append-icon offer a straightforward approach to incorporate positioned icons,

```
<v-btn append-icon="mdi-account-circle" prepend-icon="mdi-check-circle">
  <template v-slot:prepend>
    <v-icon color="success"></v-icon>
  </template>
  Button
  <template v-slot:append>
    <v-icon color="warning"></v-icon>
  </template>
</v-btn>
```

For the prepend-icon and append-icon, the corresponding slots are prepend and append

Demo

Variants

Value	Example
elevated	BUTTON
flat	BUTTON
tonal	BUTTON
outlined	BUTTON
text	BUTTON
plain	BUTTON

Containment: Cards

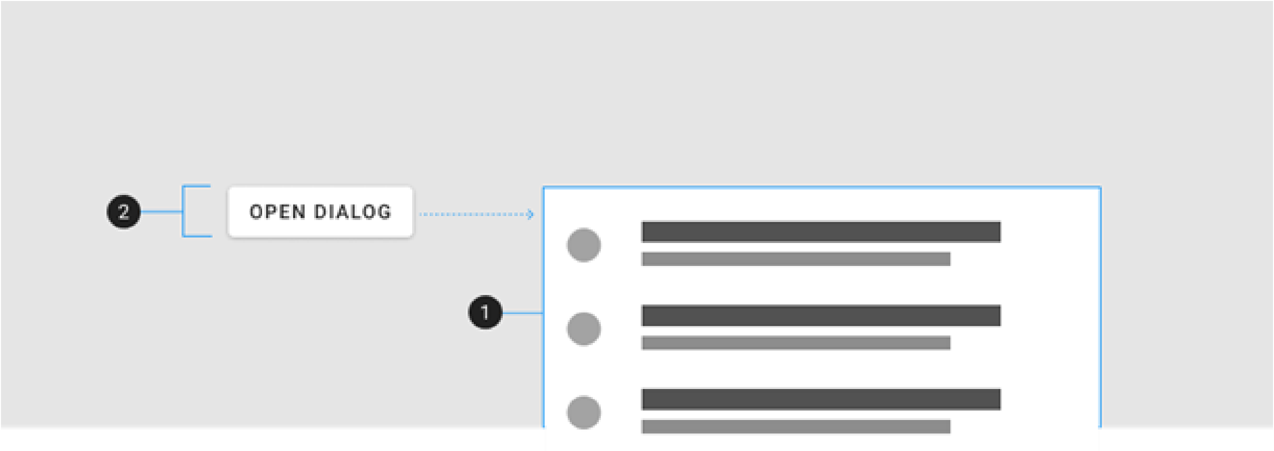
Vuetify Docs

Element / Area	Description
1. Container	The Card container holds all `v-card` components. Composed of 3 major parts: `v-card-item`, `v-card-text`, and `v-card-actions`
2. Title (optional)	A heading with increased **font-size**
3. Subtitle (optional)	A subheading with a lower emphasis text color
4. Text (optional)	A content area with a lower emphasis text color
5. Actions (optional)	A content area that typically contains one or more v-btn components

Demo

Containment: Dialogs

[Vuetify Docs]



Element / Area	Description
1. Container	The dialog's content that animates from the activator
2. Activator	The element that activates the dialog

<v-data-table>

Vuetify Docs

Title ↑	Author	Genre	Year
1984	George Orwell	Dystopian	1949
Pride and Prejudice	Jane Austen	Romance	1813
The Great Gatsby	F. Scott Fitzgerald	Tragedy	1925
To Kill a Mockingbird	Harper Lee	Southern Gothic	1960

Items per page: 10 1-4 of 4

```
<template>
  <v-app>
    <v-container>
      <v-data-table :items="books"></v-data-table>
    </v-container>
  </v-app>
</template>
```

Demo

Custom Headers for Vuetify <v-data-table>

```
<v-data-table
  :items="books"
  :headers="[
    { title: 'title', align: 'start', sortable: false, key: 'title' },
    { title: 'author', align: 'end', sortable: true, key: 'author' },
    { title: 'year', align: 'end', sortable: true, key: 'year' },
  ]"
></v-data-table>
```

[Demo]

Adding Selection to Vuetify <v-data-table>

```
<v-data-table
  :items="books"
  show-select
  :headers="[
    { title: 'title', align: 'start', sortable: false, key: 'title' },
    { title: 'author', align: 'end', sortable: true, key: 'author' },
    { title: 'year', align: 'end', sortable: true, key: 'year' },
  ]"
  v-model="selected"
  item-value="title"
></v-data-table>
```

```
import { ref } from "vue";
const selected = ref([]);
```

Demo

Adding Search to Vuetify <v-data-table>

```
<v-text-field
  v-model="search"
  append-icon="mdi-magnify"
  label="Search"
  single-line
  hide-details
></v-text-field>
<v-data-table
  :items="books"
  show-select
  :headers="[
    { title: 'title', align: 'start', sortable: false, key: 'title' },
    { title: 'author', align: 'end', sortable: true, key: 'author' },
    { title: 'year', align: 'end', sortable: true, key: 'year' },
  ]"
  v-model="selected"
  item-value="title"
  :search="search"
></v-data-table>
```

```
import { ref } from "vue";
const selected = ref([]);
const search = ref("");
```

Demo

Practice: Responsive Product Card Dashboard

Requirements

- Create v-cols for all products defined in the script.
- Each v-col should contain a v-card showing:
    - name, price, and a Buy button
- Use Vuetify's grid system to make the layout responsive:
    - 4 columns per row on large screens
    - 2 columns per row on medium screens
    - 1 column per row on small screens